

Product Requirements Document

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Summary

The rising cost-of-living and environmental concerns have left consumers frustrated by buying groceries only to throw away expired food. To solve this, we are launching an AI-powered Smart Pantry that allows users to scan their receipts, automatically estimates expiration dates, and suggests personalized, budget-friendly recipes based on what needs to be eaten first. This completely removes the need for manual data entry while actively helping our users save money and reduce waste.

Target Personas

- **Persona A (Sarah):** A 32-year-old working parent who is time-poor, budget-conscious, and needs quick 30-minute recipes her kids will eat.
- **Persona B (Liam):** A 20-year-old eco-conscious university student who hates food waste but lacks the cooking skills to invent a meal out of leftover ingredients.

User Stories

User Story 1 (For Sarah)

As a working parent on a tight budget, I want the app to suggest quick, 30-minute meals using ingredients that are about to expire, so that I can feed my kids without wasting food or stressing about dinner.

Acceptance Criteria

- The AI recipe generator must include an "Under 30 Minutes" filter that defaults to on.
- The algorithm must prioritize ingredients in the user's inventory that expire within the next 3 days.

User Story 2 (For Liam)

As a beginner cook, I want the app to scan my receipt and guess the expiration dates for my groceries, so that I do not have to manually log everything or guess when my food is going to go bad.

- The receipt scanning tool needs to correctly read and log at least 85% of the items on a standard supermarket receipt.
- The system must automatically assign a standard estimated shelf life to fresh produce as soon as it is scanned.

User Story 3 (Addressing Privacy Concerns)

As a privacy-conscious user, I want clear reassurance about how my receipt data is being used, so that I feel safe scanning my purchases without worrying my habits are being sold.

- A friendly tooltip must pop up the first time a user opens the camera, confirming their data is secure and never sold to advertisers.
- We must include a toggle in the settings menu allowing users to anonymize their purchasing data.

User Story 4 (Addressing the Cost-of-Living Crisis)

As a budget conscious shopper, I want the app to track how much money I saved by cooking food before it expired, so that I can see the financial value of using the app.

- The app must display a "Money Saved" metric that calculates the estimated value of expiring items that were marked as cooked.
- This metric must update automatically on the user's main dashboard after completing an AI-suggested recipe.

Success Metrics

Recipe Engagement

- The percentage of users who see an AI-suggested recipe and click the button to cook it.
- Because this tells us if the AI is doing a good job. If the app suggests a meal and no one ever makes it, the feature isn't really solving their problem.

Waste Reduction

- We will track if the total number of items users throw away goes down each month.
- Because this number drops over time, it proves our app is genuinely helping people waste less food and save money.

Scanner Usage

- The percentage of our active users who take a picture of their receipt at least once a week.
- Because if people stop scanning their receipts after the first try, the app won't have any groceries to track. We need to watch this to make sure the camera tool is quick and easy to use.